

A sequential disassembly planning approach based on knowledge graph and graph isomorphism network for supporting power battery remanufacturing

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ABSTRACT

Disassembly is an important step during end-of-use power battery remanufacturing that offers a desired solution to reclaim the battery function and material values into new life cycles. Battery disassembly knowledge management and relationship reasoning plays a significant role in intelligent disassembly planning. However, a disassembly system that depends on predefined product conditions or fixed disassembly relationship, is not capable of handling highly uncertain and complicated end-of-use product disassembly. To overcome the bottleneck, this paper proposes an integrated disassembly framework based on knowledge graph and Graph Isomorphism Network, which significantly enhances both disassembly information integration and reasoning capability. First, the TransH knowledge graph embedding technique is introduced to capture multiple relationships between graph nodes such as components and tools during disassembly, while Graph Isomorphism Network further strengthens the model's ability to learn the complex topological structures among parts in the battery disassembly process, ensuring accurate handling of diversity and complexity in disassembly. On this basis, the model utilized the decomposition-based Multi-Objective Evolutionary Algorithm to break down the multi-objective optimization problem into several single-objective sub-problems, generating a Pareto frontier solution set to achieve an optimal balance and trade-off among various objectives, such as disassembly cost and carbon emissions. The experiment on a retired power battery demonstrates that the systematic model excels in disassembly planning, effectively balancing conflicting objectives and improving both disassembly efficiency and resource utilization. Compared to classical algorithms like NSGA-II, SPEA2, and SMS-EMOA, the framework improves the Total Objective Value (TOV) by approximately 5 %, while solution set coverage and the proportion of non-dominated solutions increase by 9.5 % and 8 %, respectively, highlighting its broad application potential in intelligent manufacturing and resource management. In the future, this model is expected to scale to more complex large-scale task scenarios, further enhancing its ability to address multi-objective optimization problems and providing crucial support for sustainable development and the green economy.

1. Introduction

Climate change has impacted the global ecological balance, threatening economic development and quality of life worldwide (Popkova and Shi, 2022). To address this environmental crisis, nations are actively promoting the development of renewable energy sources and accelerating the transition of their energy structures. Clean energy sources such as solar, wind, and hydropower are gradually replacing fossil fuels, driving a global shift towards a low-carbon economy. This transition is not only essential for combating climate change, but also a strategic

choice to ensure energy security and achieve sustainable development (Rahman et al., 2025). As one of the world's largest energy consumers, China has demonstrated its commitment to resource management and environmental protection by implementing policies such as the "Interim Measures for the Management of Recycling of New Energy Vehicle Batteries." These policies aim to foster clean energy usage and comprehensive resource utilization, striving to establish a low-carbon, green economic system (Cerchione et al., 2025). With advancements in technology and supportive policies, China's production and consumption of clean energy have grown steadily. By 2050, renewable

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energy is expected to account for the majority of the global energy supply. In this shift, lithium-ion batteries play a crucial role as core components for clean energy storage and electric transportation (International Energy Agency (IEA), 2023). However, the widespread use of lithium-ion batteries brings severe environmental and resource challenges. By 2030, it is estimated that over one million tons of waste lithium-ion batteries will be generated annually. If improperly handled, the heavy metals and harmful substances in these batteries could cause severe environmental pollution, while the unutilized resources such as lithium, cobalt, and nickel would lead to a waste of non-renewable resources (Mrozik et al., 2021). Consequently, exploring efficient battery disassembly and recycling methods has become a critical task for achieving a circular economy and sustainable development. Existing battery recovery methods, such as mechanical and chemical processing, show clear deficiencies in terms of resource recovery rate, processing cost, and environmental impact, making it difficult to achieve an optimal balance between recovery efficiency and environmental requirements.

Remanufacturing allows the reclamation of both battery functional and material values into new life cycles by disassembling battery packs into certain levels, screening and grouping of retired modules/cells into new packs as energy storage devices in the wind/solar farms or similar scenarios. In this way, the useful life of end-of-use batteries can be extended in replace of new batteries to mitigate natural resources consumption, elements/compounds refinement and resultant pollution. Thus, disassembling the end-of-use batteries is the first and foremost step, involving the disconnection of various fasteners and joints, and removal of functional components. It is noted that the intricate structure of lithium-ion batteries presents strong inter-component dependencies. The various types and varied deterioration of connections among different battery components increase separation difficulty and uncertainty, making it challenging for fixed or deterministic approaches to effectively manage these complex dependencies, leading to unexpected disruptions, and resultant low efficiency and high cost.

Therefore, this paper proposes a multi-objective optimization method that combines TransH (Wang et al., 2014) knowledge graph embedding with the Graph Isomorphism Network (GIN) (Xu et al., 2019) to optimize an intelligent disassembly model for lithium-ion batteries, aiming to achieve efficient resource recovery. From a theoretical perspective, this paper introduces an innovative theoretical framework into the field of battery disassembly, employing knowledge graph embedding techniques to capture the complex relationships between components and tools during the disassembly process, while using Graph Neural Networks (GIN) to further enhance the model's ability to learn complex topological structures. This innovation not only fills the gap in current research regarding the handling of component dependency relationships in battery disassembly but also provides a new perspective and method for multi-objective optimization theory.

On the methodological level, this paper leverages the advantages of TransH and GIN to overcome the balance challenge between recovery efficiency, cost control, and environmental protection in traditional battery disassembly methods. By introducing multi-objective optimization algorithms, this paper effectively considers multiple objectives such as recovery rate, disassembly cost, and carbon emissions, ensuring that resource recovery efficiency is enhanced while minimizing environmental impact. The method demonstrates clear advantages in handling the complexity and diversity of the battery disassembly process, particularly in addressing the complex dependency relationships among different components within the battery.

From a management perspective, the proposed optimization model not only brings theoretical innovation but also has significant practical implications. By providing the battery recycling industry with a more efficient and intelligent disassembly path optimization method, this research contributes to improving resource recovery rates, reducing disassembly costs, and minimizing carbon emissions, offering effective support for companies in green manufacturing and sustainable

development. Furthermore, as policies promoting circular and green economies continue to evolve, the findings of this study provide strong theoretical support for policymakers, facilitating the effective implementation of environmental protection and resource management policies.

The structure of the paper is as follows: Section 2 provides an overview of related literature and contributions; Section 3 describes the overall approach and research methodology; Section 4 presents disassembly information, typical cases, and experimental results; and Section 5 offers conclusions and future research directions.

2. Related work

In the field of battery disassembly planning, numerous studies have explored various models and optimization algorithms. In contrast to traditional modelling techniques, knowledge graph is able to capture the dynamic disassembly information such as the variation of part quality and part disassembly relationships to deal with uncertainty and information reuse. Besides, updateability and extensibility can be expected for knowledge graph when the unexpected disassembly scenarios are captured by cameras in a highly intelligent disassembly system. Moreover, knowledge graph embedding can help capture complex part disassembly relationships to facilitate disassembly reasoning for optimal disassembly sequence. Chen et al. (2023) utilized the TransE method to construct a two-level prediction framework, capturing the relationships between users and content through translation embedding, thereby enhancing recommendation accuracy; Li et al. (Li and Zhu, 2024) introduced the multi-translation principle based on TransE to effectively address complex relationships within knowledge graphs; Karetnikov et al. (2022) proposed an extended TransE model applied to behavior prediction in temporal knowledge graphs, demonstrating superior predictive performance; Wan et al. (2023) enhanced the TransH model by incorporating relational neighborhood information and a flexible translation strategy, improving performance in triple classification and link prediction tasks; Wang et al. (2023) further optimized TransH by using hyperplane projection and relational attributes to process many-to-many relationships, achieving higher processing effectiveness.

In recent years, Graph Neural Networks (GNNs) have been widely used to enhance the embedding performance of knowledge graphs. He et al. (2023) proposed the Cluster TransD-GAT model, which integrates Graph Attention Network (GAT) to effectively capture relationship weights within knowledge graphs, significantly optimizing the computational efficiency of TransD in large-scale data applications; Chang et al. (2024) combined GCN with TransE to improve the entity alignment accuracy in blockchain intelligence graphs; Ma et al. (2022) proposed the RotatSAGE model, which integrates RotatE and GraphSAGE, enhancing accuracy through rotational embeddings and the augmentation of neighbor information; Xiao et al. (Xiao and Li, 2023) developed the TransH-SAGE model, which combines GraphSAGE and TransH for Chinese movie recommendations, achieving high recommendation performance; Jiang et al. (2024) introduced a socially-aware recommendation method based on Graph Attention Networks, which aggregates preference features from knowledge graphs and social graphs, further optimizing the performance of the recommendation system.

Yu et al. (2022) proposed a battery disassembly planning method based on ontology and local destruction rules, optimizing the disassembly sequence and reducing time and costs; Chu et al. (Chu and Chen, 2023) enhanced the collaborative optimization of battery disassembly scheduling and strategies by integrating Q-learning with a hybrid particle swarm optimization algorithm (HPSO-QL); Guo et al. (Guo and Zhou, 2023) designed a multi-objective dynamic programming method based on improved AND/OR graphs and the Pareto dominance concept, effectively achieving multi-objective optimization of disassembly sequences; Zhou et al. (Zhou and Wu, 2020) optimized the sequence transition time and energy consumption of assembly lines based on the MOEA/D algorithm, achieving a balance between green manufacturing

Table 1
Literature Review Summary table.

Model/Method	Main Contribution
TransE (Chen et al., 2023)	Captures user-content relationships, enhancing recommendation accuracy
TransE (Li and Zhu, 2024)	Introduces the multi-translation principle to optimize complex relationship modeling
Extended TransE (Karetnikov et al., 2022)	Applies temporal knowledge graph prediction to improve prediction performance
Enhanced TransH (Wan et al., 2023)	Introduces a flexible translation strategy to improve classification and prediction performance
Improved TransH (Wang et al., 2023)	Optimizes multi-relation handling via hyperplane projection
TransD-GAT (He et al., 2023)	Combines GAT and TransD to optimize computational efficiency
GCN + TransE (Chang et al., 2024)	Enhances entity alignment accuracy in blockchain graphs
RotatSAGE (Ma et al., 2022)	Combines RotatE and GraphSAGE to improve accuracy
TransH-SAGE (Xiao and Li, 2023)	Combines GraphSAGE and TransH to improve movie recommendation effectiveness
GAT (Jiang et al., 2024)	Aggregates preference features to optimize recommendation systems
Local Disruption (Yu et al., 2022)	Optimizes battery disassembly sequence, reducing time and cost
Q-Learning (Chu and Chen, 2023)	Improves battery disassembly scheduling optimization
Pareto Dominance (Guo and Zhou, 2023)	Optimizes disassembly sequence for multi-objective
MOEA/D (Zhou and Wu, 2020)	Optimizes assembly line sequence conversion time and energy consumption for green manufacturing
NSGA-II (Wu et al., 2022)	Provides a robot-human collaborative disassembly solution with flexible decision support

goals and productivity; Wu et al. (2022) proposed an NSGA-II model that facilitates the disassembly of waste power battery modules through the collaboration of robots and manual labor, providing flexible decision support for enterprises.

Despite the significant progress made in knowledge graph embedding, multi-objective optimization, and battery disassembly optimization, several limitations remain. First, existing knowledge graph embedding methods are predominantly focused on recommendation systems and simple relationship modeling, making it challenging to effectively apply them to the complex component dependencies and sequence optimization required in battery disassembly. Second, while Graph Neural Networks have enhanced performance in certain scenarios, they still fall short in capturing the multi-layered dependencies and dynamic complexities of battery disassembly, limiting their ability to comprehensively represent interactions between entities. Lastly, although current multi-objective optimization algorithms offer solutions for factors such as carbon emissions and tool transition times, they do not sufficiently account for practical engineering constraints. This inadequacy is particularly pronounced in addressing highly complex industrial scenarios, where it becomes difficult to effectively balance multiple optimization objectives, thereby failing to achieve systemic overall optimization.

As shown in Table 1, despite significant progress in the areas of knowledge graph embedding, multi-objective optimization, and battery disassembly optimization, there are still some limitations. First, existing knowledge graph embedding methods are mostly focused on recommendation systems and simple relationship modeling, making it difficult to effectively address the complex component dependencies and sequence optimization issues in battery disassembly. For instance, traditional models such as TransE and TransH perform poorly when dealing with higher-order relationships and multi-layered dependencies, making them difficult to apply in complex industrial scenarios. Although extended models like TransE and TransH have optimized relationship modeling to some extent, they still fail to adequately handle multi-dimensional complex relationships.

Secondly, while Graph Neural Networks (GNNs) have made progress

in improving the performance of recommendation systems, they still have limitations when dealing with multi-layered dependency relationships and dynamic evolution in battery disassembly. Models like TransH-SAGE and RotatSAGE improve accuracy by combining GNNs with embedding models, but they heavily depend on neighbor information and suffer from instability in complex scenarios, making it challenging to fully represent the interactions between entities.

Finally, although existing multi-objective optimization algorithms offer solutions for optimizing carbon emissions, tool conversion times, and other factors, their adaptability to practical engineering constraints remains insufficient. For example, algorithms such as Q-Learning combined with Particle Swarm Optimization (HPSO_QL) and NSGA-II improve disassembly scheduling, but still face long computation times in high-dimensional complex scenarios and struggle to balance multiple optimization objectives effectively.

3. Overall framework

To address the aforementioned limitations, as shown in Fig. 1, this paper proposes the THG-MOEA/D framework. It combines knowledge graph embedding (TransH), Graph Isomorphism Networks (GIN), and multi-objective optimization algorithm (MOEA/D) (Zhang and Li, 2007) methods to enhance the efficiency and resource utilization in battery disassembly. The framework consists of the disassembly operation data processing module, the TransH-GIN modeling module, the multi-objective optimization module, and the optimal disassembly sequence selection module. These modules work synergistically, integrating knowledge graph embedding with multi-objective optimization techniques, aiming to improve resource recovery rates and promote environmental sustainability.

The core of the framework lies in utilizing knowledge graph embedding techniques to construct the dependency relationships between components in the disassembly path, thereby providing data support for the subsequent optimization process. The TransH and GIN models capture the complex dependencies and topological structures within the disassembly process, ensuring that the optimization model can effectively balance objectives such as recovery rate, cost, and environmental impact. Finally, based on the evolutionary multi-objective optimization algorithm MOEA/D, the framework is able to find a balance between multiple optimization objectives and select the optimal disassembly path.

3.1. Knowledge graph construction

As shown in Fig. 2, the disassembly operation data processing is responsible for handling the diverse and heterogeneous data involved in the battery disassembly process, including battery components, disassembly time, connection methods, and disassembly tools. To effectively manage this information, the framework adopts entity-relationship extraction techniques to identify key entities and their relationships from the disassembly steps. A knowledge graph is constructed to represent the dependencies and structured information between entities. The construction of the knowledge graph forms the foundation of the entire framework, where each node represents an entity (such as a battery component), and each edge denotes the relationship between entities (such as connection methods or disassembly order). This enables the structured processing of data and provides the topological structure and dependencies for subsequent modeling, ensuring the rationality of the disassembly path selection.

3.2. Graph embedding – TransH

The obtained knowledge graph of battery disassembly is embedded into a low-dimensional vector space by mapping entities and relationships, enabling subsequent tasks. TransE (Bordes et al., 2013) is one of the pioneering approaches in knowledge graph embedding, inspired by

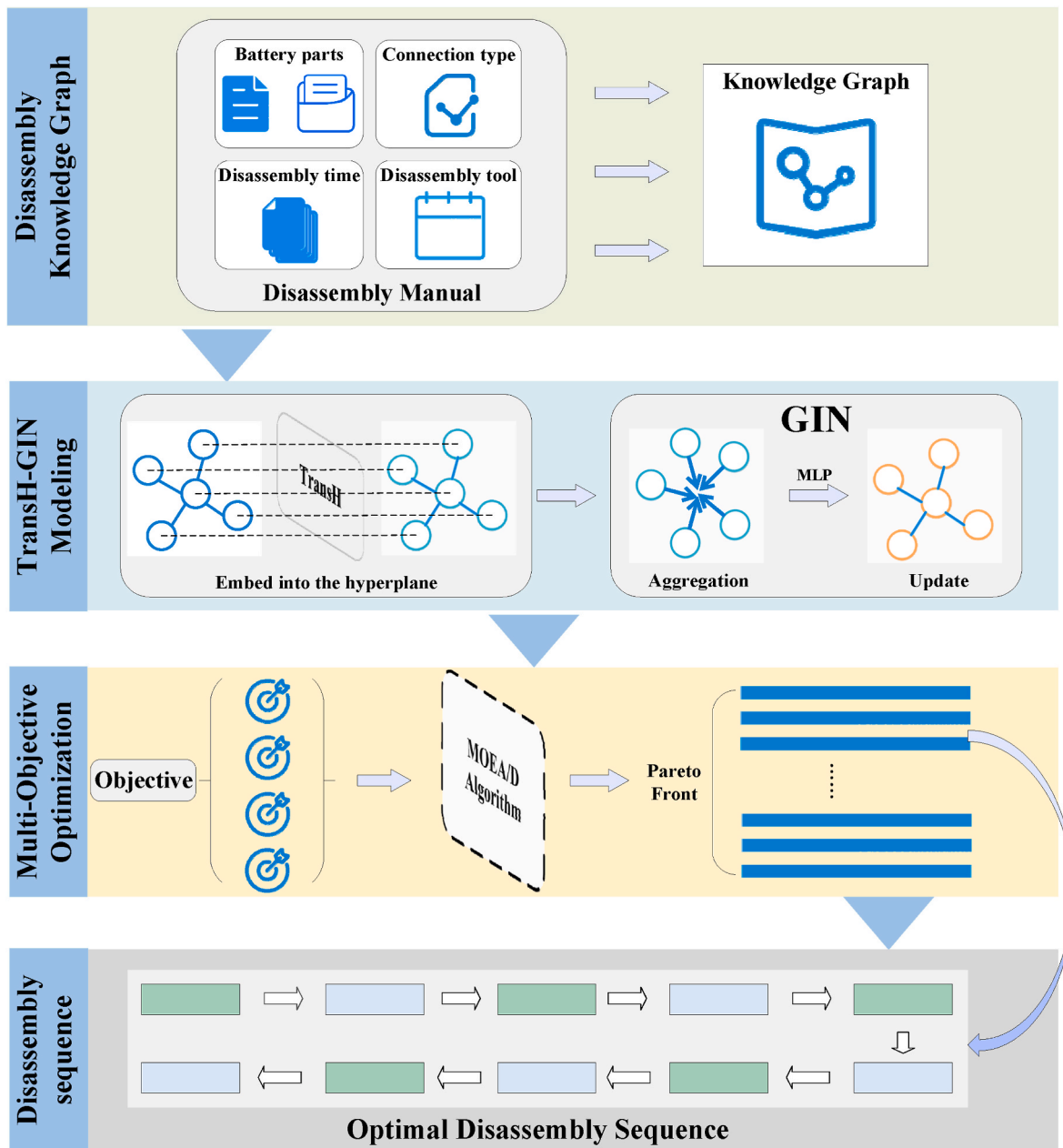


Fig. 1. Overview of the THG-MOEAD Model Framework. The four modules of the entire process, are arranged from top to bottom, emphasizing the key components of the framework.

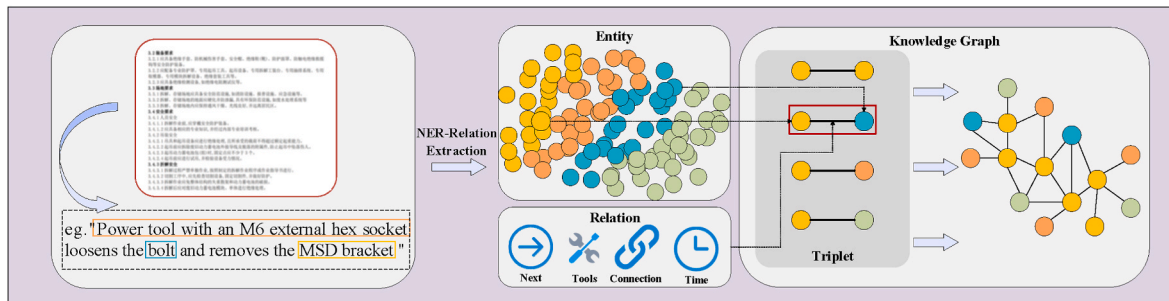


Fig. 2. Knowledge Graph Construction Diagram. Constructing the knowledge graph is presented from left to right, where entities of different colors represent different categories, and edges denote the dependencies between entities.

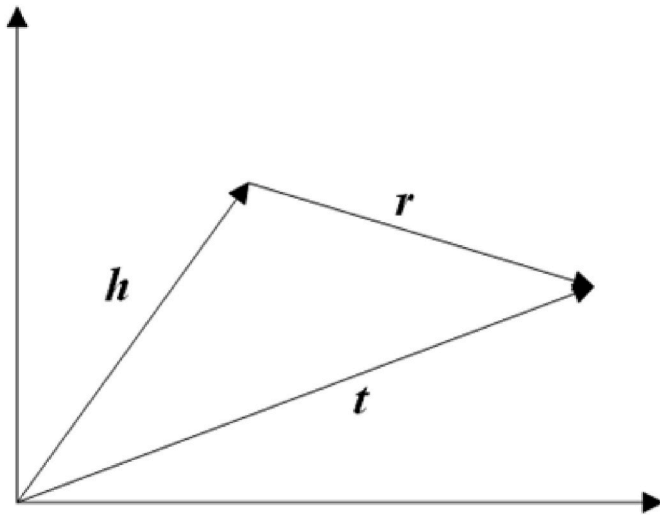


Fig. 3. Structure diagram of the TransE model.

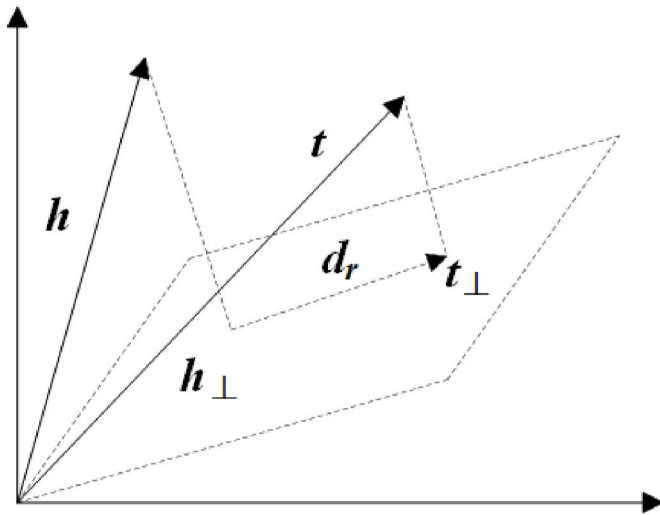


Fig. 4. Schematic diagram of the TransH model structure.

the Word2vec(Mikolov et al., 2013) word embedding model, aiming to represent entities and relationships in knowledge graphs in a simple yet effective manner. As shown in Fig. 3, it treats relationships as translation operations in a low-dimensional embedding space. For any given triple (h, r, t) , the entities and relationships are first mapped to a d -dimensional low-dimensional space, resulting in entity vectors (h, t) and relationship

vectors r in the same space. If the triple is correct, the head entity vector plus the relationship vector should approximate the tail entity vector, that is, $h + r \approx t$.

When handling complex battery disassembly tasks (such as dealing with battery components connected in various ways), TransE may exhibit limitations, especially when addressing one-to-many or many-to-many relationships. Therefore, the TransH model was further developed based on TransE. TransH employs the concept of projection, allowing entities to have different representations under specific relationship types, thus enhancing the model's ability to handle complex relationships. In the context of battery disassembly, this improvement enables a more precise representation of the relationships between different components in the battery. For instance, the connection method of a battery module may involve multiple disassembly steps. TransH constructs corresponding hyperplanes (as shown in Fig. 4) for specific relationships and projects the entities onto these hyperplanes, allowing disassembly paths to be more clearly represented, thereby improving optimization efficiency. As illustrated in Fig. 4, a corresponding hyperplane is established for specific relationships, modeling the normal vector n_r of the hyperplane and the relationship vector d_r . The entity vectors are then projected along the normal vector of the plane onto the hyperplane, where the relationship vector d_r acts as a transformation operation for the projected entity vector on the hyperplane. The projected entity vectors represent the translations of the head and tail entities on the hyperplane, thus better capturing the characteristics of complex relationships.

The projection formula is shown in Equation (1):

$$h' = h - (n_r^T h) n_r$$

$$t' = t - (n_r^T t) n_r \quad (1)$$

TransH defines the effectiveness of relationships by calculating the distance between the mapped entities, typically using the L2 distance for measurement. The loss function is in the form shown in Equation (2):

$$L = \sum_{(h,r,t) \in \text{positive}} \max(0, d(h', t') + \gamma - d(h', t_{\text{neg}})) \quad (2)$$

The TransH model is used to capture the relationships between different components during the battery disassembly process, mapping them through hyperplanes to obtain corresponding embedding vector representations. This enables the handling of various types of relationships. The model efficiently distinguishes the impact of different connection methods on entities, providing structured input for the subsequent optimization process.

3.3. Graph isomorphism network

Graph Isomorphism Network (GIN) represents a significant advancement in deep learning models for graph data processing in

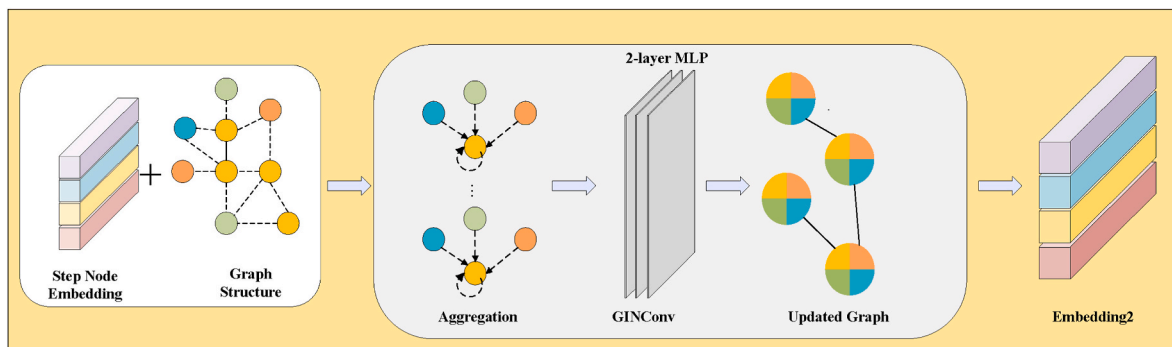


Fig. 5. GIN Architecture Diagram. The diagram represents the flow from input to output through GIN, where the 2-layer MLP structure consists of the first linear layer, a ReLU activation function, and the second linear layer.

recent years. Initially, research on Graph Neural Networks (GNNs) primarily focused on efficiently handling graph-structured data. Early GNN models, such as Neural Message Passing (NMP) (Gilmer et al., 2017) and Graph Convolution Network (GCN) (Kipf and Welling, 2017), updated node representations by aggregating features from neighboring nodes. However, these models faced limitations in addressing complex graph isomorphism issues, particularly in distinguishing between different graph structures.

As research on graph isomorphism progressed, it became evident that traditional feature aggregation methods were insufficient to capture the complexity of graphs, leading to the development of more robust models. GIN emerged in this context, inspired by graph isomorphism theory, aiming to enhance the representation power of graph structures through a refined node aggregation mechanism.

The core idea of GIN is to update the representations of nodes by aggregating information from the nodes and their neighbors. Compared to previous models, GIN specifically introduces a learnable parameter, allowing for more flexible node updates under specific relationships. In GIN, the node representation update formula is shown in Equation (3):

$$h_v^{(k)} = \text{MLP}^{(k)} \left((1 + \varepsilon) h_v^{(k-1)} + \sum_{u \in N(v)} h_u^{(k-1)} \right) \quad (3)$$

Where $h_v^{(k)}$ is the representation of node v at the k -th layer, $N(v)$ is the set of neighbors of node v , $\text{MLP}^{(k)}$ is the multi-layer perceptron used for node updates, and ε is a learnable parameter.

The structure of GIN is shown in Fig. 5. GIN focuses on local topological information and aggregates the features of neighboring nodes, effectively capturing the complex dependency relationships between different component nodes in the battery. By taking node features (Step Node Embedding) and the graph's topology (Graph Structure) as inputs, it performs an aggregation operation to obtain initial node representations. Then, through fully connected layers (MLP) (Kipf and Welling, 2017) and ReLU activation, the features are updated to generate the final node embedding representations (Embedding2). This approach allows GIN to demonstrate strong flexibility and expressive power when handling disassembly paths, effectively learning the embedding representations of entity nodes and ensuring the proper handling of complex dependency relationships.

3.4. The MOEA/D algorithm

The core idea of MOEA/D (Multi-Objective Evolutionary Algorithm based on Decomposition) is to decompose a multi-objective optimization problem into multiple single-objective subproblems for simultaneous solving. Through collaborative optimization based on neighborhood relationships, these subproblems progress together, ultimately obtaining a Pareto front solution set. MOEA/D effectively balances disassembly objectives such as disassembly cost, and carbon emissions, optimizing the disassembly path to improve resource recovery efficiency.

3.4.1. Disassembly objectives

In the optimization algorithm proposed in this paper, four disassembly objectives are simultaneously considered and integrated into the MOEA/D algorithm framework for comprehensive optimization. Specifically, the MOEA/D framework converts these objectives into optimization functions and gradually adjusts the disassembly path to achieve a balance between the objectives.

- (1) Operational Cost, this is a composite factor that includes tool conversion time, distance, and disassembly difficulty, which is minimized during the optimization process. The algorithm automatically adjusts the disassembly path to reduce unnecessary tool conversions and steps, thereby lowering the operational cost.

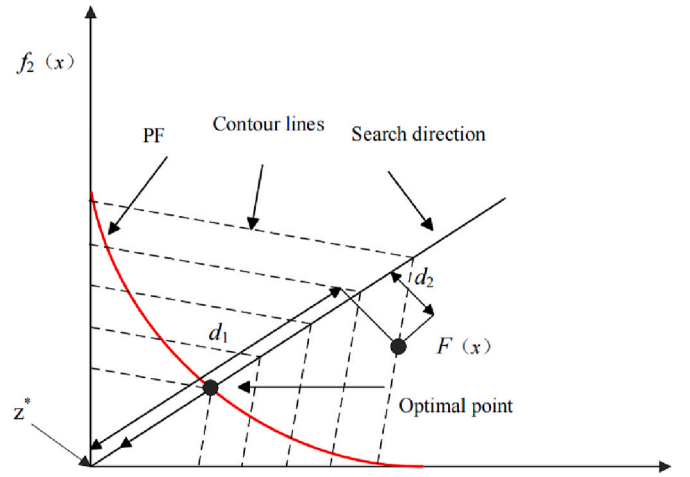


Fig. 6. Illustration of the PBI decomposition mechanism.

- (2) Carbon Emissions, during optimization, the algorithm aims to select disassembly paths that produce lower carbon emissions. The carbon emissions for each operation step are calculated and included in the objective function, ensuring that the final disassembly path meets environmental sustainability requirements.
- (3) Disassembly Reliability, by calculating the similarity between disassembly paths, the optimization algorithm tends to select those paths with higher reliability. The algorithm evaluates the similarity of different paths and prioritizes those with high reliability and low risk.
- (4) Disassembly Risk, the algorithm adjusts the disassembly path based on potential risk factors during the process, aiming to reduce the possibility of operational errors. By assessing the risk of each operation step, the algorithm minimizes the chances of unexpected events during disassembly.

By comprehensively considering these objectives, the MOEA/D framework can find an optimal disassembly path that maximizes resource recovery rate while minimizing operational costs, carbon emissions, risks, and improving reliability.

3.4.2. MOEA/D algorithm flow

In this study, the MOEA/D algorithm representation uses the Penalty Boundary Intersection (PBI) method to decompose the MOP into N sub-optimization problems. The decomposition mechanism is shown in Fig. 6, and its mathematical formulation is given in Equation (4):

$$\text{minimize } g^{\text{pbi}}(x|\lambda, z^*) = \max_{1 \leq i \leq m} \{ \lambda_i |f_i(x) - z^*| \} \quad (4)$$

Where z^* is the reference point, and $\lambda_i = (\lambda_1, \lambda_2, \dots, \lambda_m)$ is the weight vector that is uniformly distributed in the objective space for the current subproblem. d_1 is the shortest distance from any point $F(x)$ on the objective function to the reference point z^* along the direction line. d_2 is the perpendicular distance from any point to the reference line. The smaller the value of $F(x)$, the smaller d_1 becomes; the smaller the perpendicular distance of $F(x)$ to the direction line, the smaller becomes. Therefore, the final objective function value becomes smaller. When the settings are appropriate, the MOEA/D with the PBI method can handle MOP problems with any shape of the Pareto front (PF).

In the MOEA/D algorithm, the neighborhood of each subproblem is defined by calculating the Euclidean distances between the T weight vectors that are adjacent to λ_i . This neighborhood partitioning is crucial for optimizing the subproblems. The algorithm then uses simulated binary crossover and polynomial mutation strategies to iteratively optimize the initial population, facilitating the collaborative evolution of the overall population through the optimization of subproblems within the

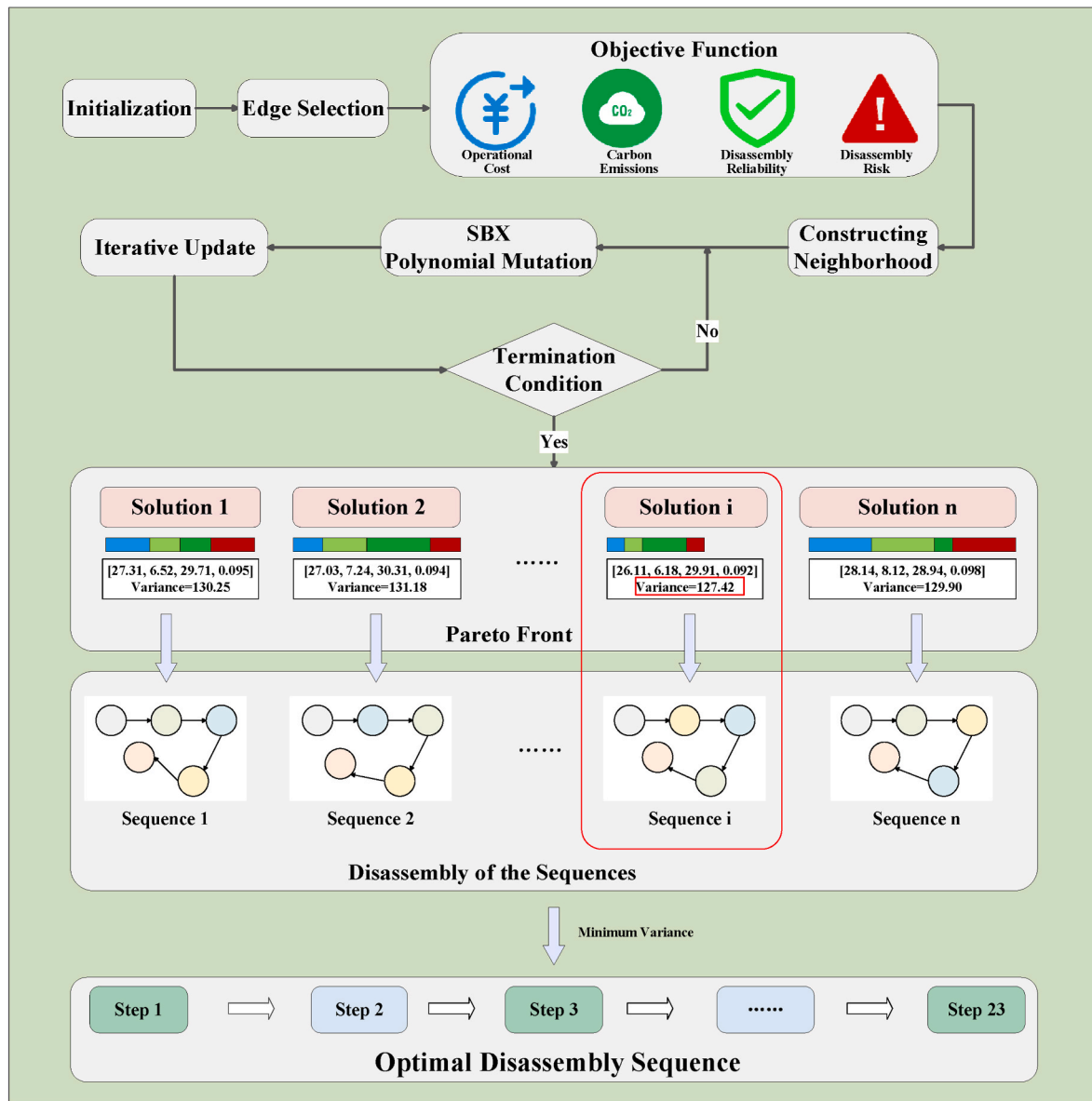


Fig. 7. MOEA/D Implementation Diagram. Where each solution represents a point on the Pareto front, reflecting its optimization performance across the four objective functions.

neighborhood.

The process of the MOEA/D algorithm based on the PBI decomposition strategy is shown as follows.

Algorithm. MOEA/D Algorithm

Input:

- MOP;
- Maximum number of iterations $MaxIter$;
- Population size N ;
- Neighborhood size T ;

Output:

Archive Set EP

Step1 Initialization

Step1.1 Set the neighborhood size T ; initialize the population size N ; external archive $EP = \emptyset$; initialize N uniformly distributed weight vectors $\lambda = \{\lambda_1, \lambda_2, \dots, \lambda_N\}$;

Step1.2 After calculating the Euclidean distance, obtain the T weight vectors closest to the weight vector λ_i , and denote the index set as $B(i)$, where $(i = 1, 2, \dots, N)$;

Step1.3 Randomly generate the initial population $X = \{x^1, x^2, \dots, x^N\}$, in the feasible space and calculate: $f(X^i), (i = 1, 2, \dots, N)$;

Step1.4 Calculate the Initialization Reference Point:

(continued on next column)

(continued)

$$z^* = (z_1^*, \dots, z_m^*), z_j^* = \min_{1 \leq i \leq N} f_j(x^i), (j = 1, 2, \dots, m);$$

Step2 Update

Step2.1 Recombination Randomly select two indices k and l from $B(i)$ that are not used for i , and perform genetic operations, including simulated binary crossover and polynomial mutation, on x^k and x^l to generate a child solution x^{child} ;

Step2.2 Update the reference point according to the following formula:

$$z^* = \begin{cases} f_j(x^{child}), & \text{if } (z_j^* > f_j(x^{child})) \\ z_j^*, & \text{otherwise} \end{cases}$$

Where $j = 1, 2, \dots, m$;

Step2.3 Update Neighboring Solutions: $\forall j \in B(i)$, if

$$g^{pbi}(x^{child} | w^j, z^*) \leq g^{pbi}(x^j | w^j, z^*),$$

Then set

$$x^j = x^{child}, f(x^j) = f(x^{child})$$

Step2.4 Update Archive: Remove all vectors in EP that are dominated by x^{child} and if none of the vectors in EP dominate x^{child} , then set $EP = EP \cup \{f(x^{child})\}$

Step3 If the stopping condition is met, then output EP ; otherwise, proceed to Step 2.

As shown in Fig. 7, the diagram illustrates the application process of the Multi-Objective Evolutionary Algorithm (MOEA/D) in solving the

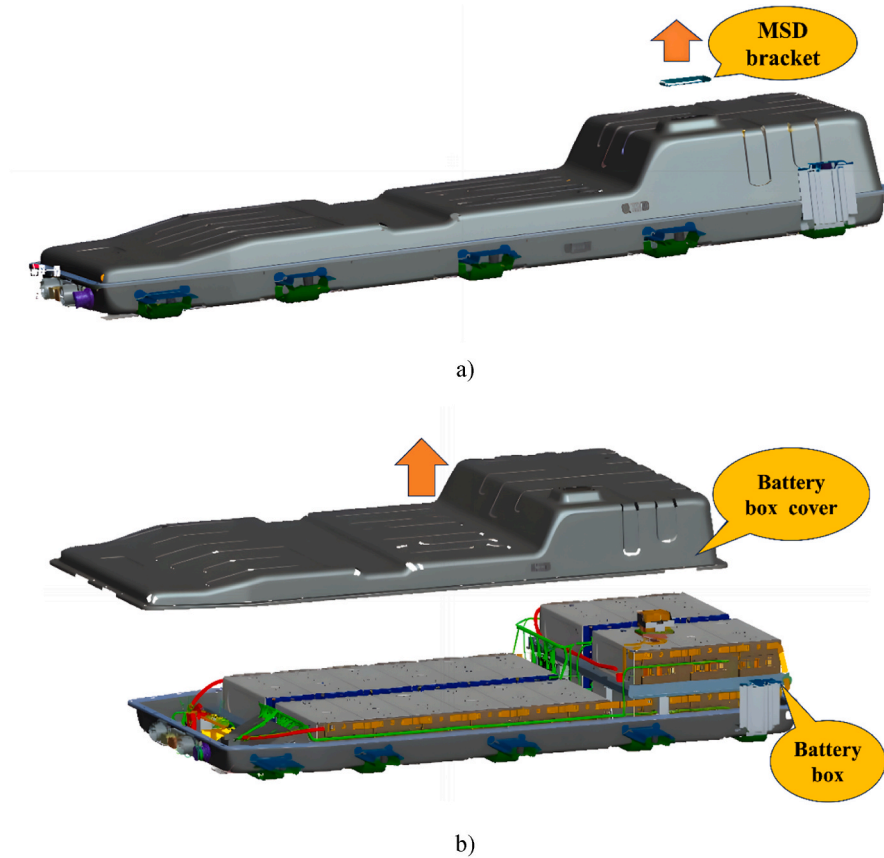


Fig. 8. Power Battery Pack shows the structure of the power battery pack, highlighting its key components and their functions in the overall system.

optimal disassembly sequence. First, the algorithm determines the task nodes and their topological relationships during the initialization phase, and sets four optimization objectives: operational cost, carbon emissions, disassembly reliability, and disassembly risk. Then, through edge selection and neighborhood construction mechanisms, combined with the SBX polynomial mutation method, the algorithm continuously explores and updates the solution space. During the iteration process, the algorithm evaluates and evolves candidate disassembly sequences based on multiple objectives, gradually forming a solution set that represents different trade-offs, known as the Pareto front. Each solution corresponds to a disassembly path and its four-dimensional objective function values, with balance assessed through variance. Finally, the solution with the minimum variance is selected from the Pareto solution set as the optimal disassembly sequence, achieving coordinated optimization among the multiple objectives and improving the efficiency and sustainability of complex product disassembly.

4. Case study of disassembly sequence analysis

To validate the effectiveness of the multi-objective optimization method based on knowledge graph embedding and Graph Isomorphism Network in battery disassembly sequencing, this study focuses on the disassembly optimization of lithium-ion power batteries from electric vehicles. These batteries have a complex structure (as shown in Fig. 8), and the disassembly process not only requires specialized tools but also involves significant safety risks.

By classifying power batteries and utilizing their structural information, a knowledge graph representing disassembly information for power batteries is generated. The Neo4j (Webber, 2012) software is used to visualize the parts of the disassembled battery and their respective connection types, as shown in Fig. 9. Partial view.

4.1. Introduction

By collecting and analyzing the obtained information, disassembly times and related details of the battery are obtained, as shown in Table 2.

Each edge connecting the components also displays the corresponding edge attribute information (the number represents the edge), including tool transition time, disassembly difficulty, carbon emissions, and so on. Table 3 shows only a portion of this information.

4.2. Experimental setup and parameter configuration

The experiment was conducted in a Windows 11 operating system environment, with hardware configured as an Intel® Core™ i5-12500H processor (2.50 GHz) and 16 GB of RAM. The development environment used the PyCharm integrated development tool, with Python 3.8 as the programming language. To support deep learning tasks, the experiment was built on the PyTorch 2.0.0 framework. This setup enabled a comprehensive performance test of the proposed model, focusing on its accuracy and robustness in complex tasks to validate its effectiveness in real-world application scenarios.

The hyperparameter configuration of the THG-MOEA/D model is shown in Table 4. The settings are based on the theoretical foundations of multi-objective evolutionary algorithms and take into account the practical requirements for model stability and solution quality in this study. The optimizer chosen is Adam, with a learning rate set to 0.001. This is selected for its ability to adaptively adjust gradients dynamically, which improves the robustness and convergence efficiency of the training process (Xu et al., 2019). The number of training epochs is set to 200, which allows effective learning while controlling overfitting risks and computational overhead (Ma et al., 2022). The number of reference directions for MOEA/D is set to 70, based on the decomposition strategy

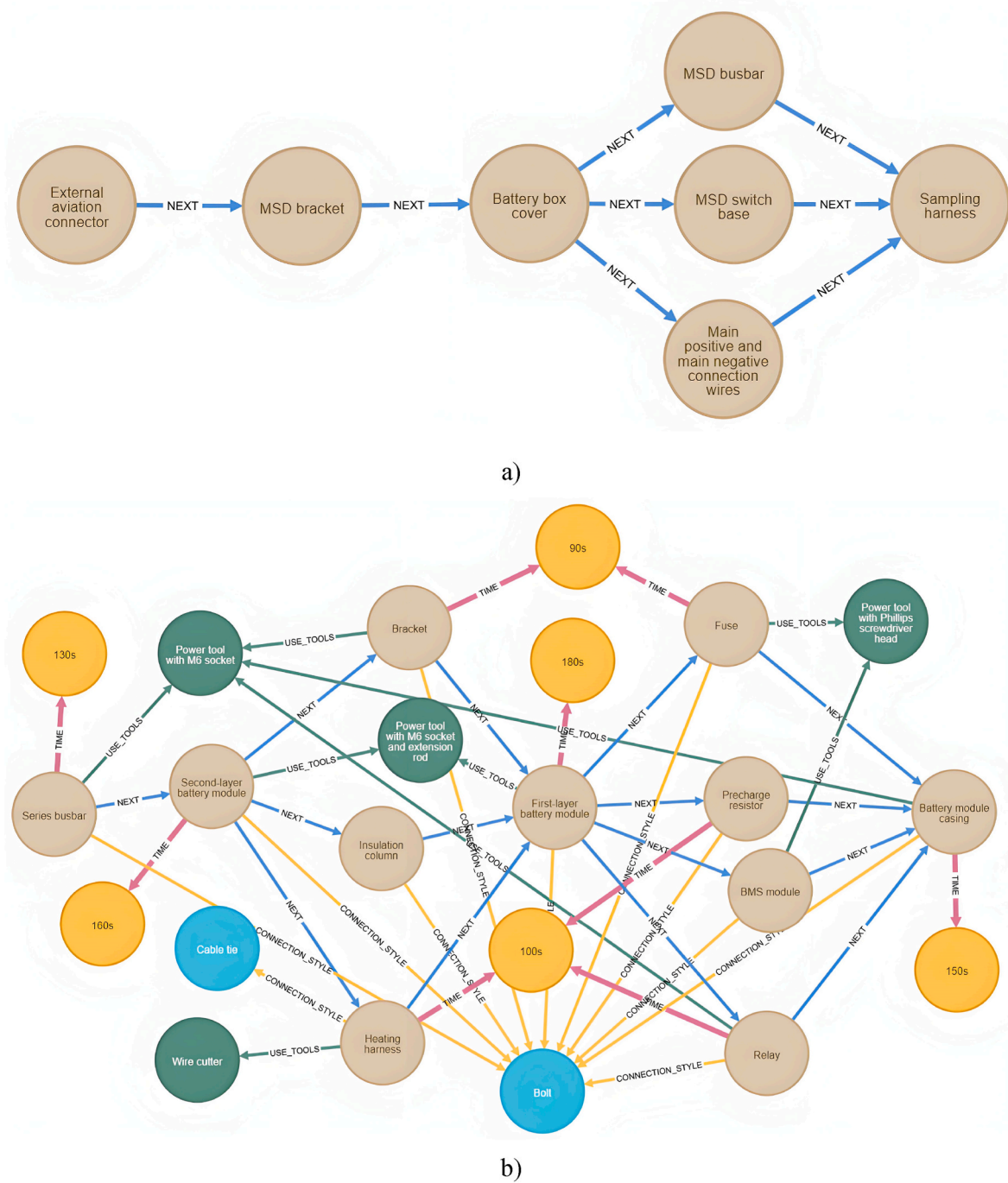


Fig. 9. Neo4j Visualization displays part of the graph, where brown nodes represent part nodes, green nodes represent tool nodes, blue nodes represent connection nodes, and yellow nodes represent time nodes. Figure a) shows the "NEXT" relationship between step nodes, while Figure b) displays the "USE_TOOLS" relationship, "CONNECTION_STYLE" relationship, and "TIME" relationship between nodes and step nodes.

proposed by Zhang and Li. This enhances the uniformity and coverage of the solution set in the objective space without significantly increasing computational burdens (Bordes et al., 2013). The number of neighbors is set to 15, with a neighbor crossover probability of 0.7, which helps to improve the quality of local search and maintain population diversity (Beume et al., 2007). In the SBX crossover operation, the crossover probability is set to 0.9, and the distribution index is set to 15, enhancing the distribution characteristics and search efficiency of the offspring solutions (Webber, 2012). Polynomial mutation uses a probability of $1/n_{var}$ and a distribution index of 20, which helps maintain population diversity and suppress premature convergence (Deb and Jain, 2014).

The maximum number of optimization iterations is set to 300 to ensure a stable and balanced non-dominated solution set in a high-dimensional objective space. Overall, the hyperparameter configuration is theoretically grounded and demonstrates good adaptability in practice, ensuring the optimization performance of THG-MOEA/D in multi-objective disassembly tasks.

4.3. Evaluation metrics

Selecting appropriate evaluation metrics is essential for a comprehensive understanding of the strengths and weaknesses of the system

Table 2
Component information table.

Number	Part Name	Conversion	Disassembly Tool	Connection
1	External aviation connector	90s	Insulating tape	Plug
2	MSD bracket	110s	Power tool with M6 external hex socket	Bolt
3	Battery box cover	120s	Power tool with M6 external hex socket	Bolt
4	MSD busbar	100s	Power tool with M6 socket	Bolt
5	MSD switch base	100s	Power tool with M6 socket	Bolt
6	Main positive and main negative connection wires	130s	Power tool with M6 socket	Bolt
7	Sampling harness	90s	Wire cutter	Cable tie
8	Series busbar	130s	Power tool with M6 socket	Bolt
9	Second-layer battery module	160s	Power tool with M6 socket and extension rod	Bolt
10	Heating harness	100s	Wire cutter	Cable tie
11	Bracket	90s	Power tool with M6 socket	Bolt
12	Insulation column	110s	Wrench	Bolt
13	First-layer battery module	180s	Power tool with M6 socket and extension rod	Bolt
14	High-voltage control system busbar	120s	Power tool with M6 socket and Phillips screwdriver	Bolt
15	BMS module	120s	Power tool with Phillips screwdriver head	Bolt
16	Relay	90s	Power tool with M6 socket	Bolt
17	Fuse	90s	Power tool with Phillips screwdriver head	Bolt
18	Precharge resistor	100s	Power tool with M4 socket	Bolt
19	Battery module casing	150s	Power tool with M6 socket	Bolt
20	Plastic partition	90s	Insulated tool	Plug
21	EVA foam	80s	Insulated tool	Plug
22	Aluminum flow guide	130s	Insulated tool	Plug
23	Battery cell	250s	Insulated tool	Plug

Table 3
Edge attribute table.

Edge	Difficulty (1–10)	Distance (mm)	Tool Transition Time (s)	Carbon Emissions (kg)
(1,2)	3	500	20	1.01
(2,3)	3	300	0	0.01
(3,4)	5	650	15	0.77
(3,5)	2	600	15	0.77
(3,6)	5	700	15	0.78
(4,7)	4	300	10	0.51
(5,7)	4	400	10	0.51
(6,7)	5	500	10	0.51
(7,8)	5	200	8	0.41

Table 4
Hyperparameter settings for model training.

Parameter Name	Value/Description	Remarks
Optimizer	Adam	Used for optimizing TransH and GIN
Learning Rate	0.001	Step size for updating model parameters
Epochs	200	Maximum number of iterations
MOEA/D Reference Directions	70	Number of reference directions for multi-objective optimization
Number of Neighbors (n_neighbors)	15	Number of neighbors in the MOEA/D
Neighbor Crossover Probability	0.7	Probability of crossover among neighbor solutions
SBX Crossover Probability	0.9	Probability of SBX crossover operation
SBX Eta	15	Distribution index in SBX crossover
Polynomial Mutation Probability	1/n_var	Probability of mutation (reciprocal of variable count)
Polynomial Mutation Eta	20	Distribution index for mutation
Maximum Optimization Iterations	300	Maximum optimization steps for MOEAD

during model performance assessment. This study employs a series of commonly used evaluation criteria for multi-objective optimization problems, covering aspects such as optimal solution selection, overall performance, computational complexity, and solution set quality. These metrics enable a thorough analysis of the algorithm's ability to balance different objectives, its computational efficiency, and the quality of the solution set, thus providing a more accurate evaluation of the model's overall performance.

(1) Optimal Sequence

In multi-objective optimization, the optimal sequence refers to a set of solutions found by the optimization algorithm that perform best across all objective functions. Given that conflicts may exist between different objectives, the Pareto frontier approach is used in the optimization process to generate a non-dominated solution set that achieves a trade-off among multiple objectives. From this Pareto frontier solution set, the solution with the minimum variance is selected as the final optimal sequence, representing the best balance across all objectives. The formula is shown in Equation (5):

$$S_{optimal} = \underset{S}{\operatorname{argmin}} \left(\sum_{i=1}^n w_i \cdot f_i(S) \right) \quad (5)$$

Where $S_{optimal}$ represents the optimal sequence, $f_i(S)$ is the value of the objective function f_i at solution S , w_i is the weight for the objective function f_i and n is the total number of objective functions. By assigning weights to each objective and measuring the variance of the solution set, multiple performance indicators can be integrated, ultimately selecting an optimal solution.

(2) Total Objective Value

The total objective value method optimizes multiple objectives by consolidating them into a single objective. By assigning weights to each objective function, multiple objectives are linearly combined into a total objective value. In this study, the weights are determined based on the solution with the minimum variance in the Pareto frontier solution set and are used to perform a weighted summation of each objective function in the optimal sequence, thereby calculating the final total objective value. The formula is shown in Equation (6):

$$TOV = \sum_{i=1}^M w_i \cdot f_i \quad (6)$$

Where TOV represents the total objective value, f_i is the value of the i -th objective function, w_i is the weight for that objective, and M is the total number of objectives. This method combines the values of multiple objectives into a scalar through a linear combination, facilitating the optimization process.

(3) Runtime (in seconds)

Runtime measures the time taken by an algorithm from execution to task completion, serving as an important metric for evaluating computational complexity and efficiency. A shorter runtime typically indicates lower time complexity or more efficient use of computational resources. Runtime directly affects the feasibility of an algorithm in real-time tasks or large-scale data processing. The formula is shown in Equation (7):

$$RT = T_{end} - T_{start} \quad (7)$$

Where T_{end} is the time when the algorithm finishes running, and T_{start} is the time when the algorithm begins execution. The difference between these two times is the total runtime of the algorithm.

(4) Model Parameter Count

The number of model parameters reflects the complexity of the model, especially in deep learning and neural networks. A larger number of parameters typically enhances the model's representational capacity but also comes with higher computational and storage costs, as well as an increased risk of overfitting. The parameter count is often used to measure the model's complexity to evaluate its computational burden and generalization ability. The formula for calculating the number of model parameters is shown in Equation (8):

$$PC = \sum_{l=1}^L (n_l \times n_{l-1} + n_l) \quad (8)$$

Where L represents the number of layers in the model, n_l is the number of neurons in the l -th layer, and n_{l-1} is the number of neurons in the $(l-1)$ -th layer. The formula calculates the sum of the parameter counts for the weight matrix and the bias term of each layer.

(5) Non-dominated Solutions Ratio

The number of model parameters reflects the complexity of the model, especially in deep learning and neural networks. More parameters often enhance the model's representational capacity but also increase computational and storage costs, along with the risk of overfitting. Parameter count is commonly used to assess model complexity, providing insights into its computational load and generalization ability. The calculation formula is shown in Equation (9):

$$NDSR = \frac{|S_{non-dominated}|}{|S|} \quad (9)$$

Where $|S_{non-dominated}|$ represents the number of non-dominated solutions, and $|S|$ is the total number of solutions in the set. This ratio indicates the percentage of non-dominated solutions in the solution set, representing the overall quality of the solutions.

(6) Hypervolume

Hypervolume is a commonly used metric in multi-objective optimization, measuring the volume covered by the non-dominated solution set in the objective space. It reflects the balance and diversity of the solution set across multiple objectives. A larger hypervolume indicates broader coverage of the objective space by the non-dominated solution set, signifying better trade-offs and performance. Therefore, hypervolume is a crucial metric for evaluating the performance of multi-

Table 5

Mean and confidence interval table.

Metric	Mean	95 % Confidence Interval
Total Objective Value	15.10	[14.32, 15.89]
Runtime (seconds)	57.32	[50.18, 64.47]
Hypervolume (HV)	879.76	[709.74, 1049.78]
Non-dominated Ratio	0.0371	[0.0337, 0.0405]

Table 6

Results presentation.

Metric	Value
Optimal Sequence	(1, 2, 3, 6, 5, 4, 7, 8, 9, 12, 11, 10, 13, 14, 18, 17, 16, 15, 19, 22, 20, 21, 23)
Total Objective Value	14.44
Runtime/s	58.46
Model Parameter Count	960
Non-dominated Solutions Ratio	0.042
Hypervolume	928

objective optimization algorithms. The hypervolume is calculated based on a reference point R_{ref} , as shown in Equation (10):

$$HV = \int_{R_{ref}} \mathbb{I}(x \prec S_{non-dominated}) dx \quad (10)$$

Where $\mathbb{I}$ is the indicator function, indicating whether the solution x is located in front of the non-dominated solution set, and HV represents the hypervolume covered by the solution set in the objective space. This

Table 7

Algorithm effectiveness.

Part Name	Minimum Operational Cost	Maximum Reliability	Optimization Algorithm Balance
External aviation connector	1	1	1
MSD bracket	2	2	2
Battery box cover	3	3	3
MSD busbar	5	4	6
MSD switch base	6	5	4
Main positive and main negative connection wires	4	6	5
Sampling harness	7	7	7
Series busbar	8	8	8
Second-layer battery module	9	9	9
Heating harness	11	11	12
Bracket	12	12	11
Insulation column	10	10	10
First-layer battery module	13	13	13
High-voltage control system busbar	14	14	14
BMS module	17	15	18
Relay	18	18	17
Fuse	15	16	16
Precharge resistor	16	17	15
Battery module casing	19	19	19
Plastic partition	21	22	21
EVA foam	22	20	22
Aluminum flow guide	20	21	20
Battery cell	23	23	23

Summary Metrics

Operational cost	26.1		
Reliability		29.91	
Total objective value			14.44

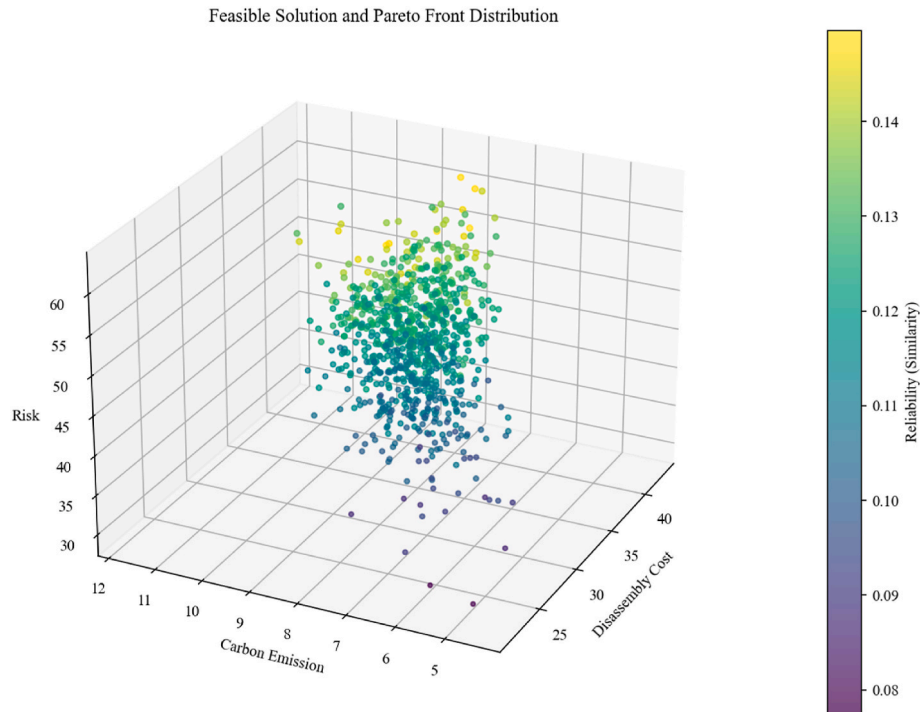


Fig. 10. The normalized distribution of feasible solutions and Pareto frontier shows the distribution of feasible solutions within the optimization space, with the Pareto frontier representing the set of optimal solutions that balance trade-offs between multiple objectives.

metric calculates the volume difference between the non-dominated solution set and the reference point.

4.4. Experimental results and analysis

4.4.1. Display of experimental results

In the experiments, by running the model multiple times (5 experiments), we calculated the mean and 95 % confidence interval for each performance metric. The specific results are shown in Table 5. The confidence interval represents the possible range for each metric, indicating that 95 % of the experimental results will fall within this range. These results provide us with a deeper understanding of the model's performance and stability in practical applications.

Through data optimization, the final results for various performance metrics were obtained, as shown in Table 6. The multi-objective optimization algorithm (MOEA/D) enabled the THG-MOEA/D model to achieve a good balance between multiple objectives and find the optimal solution with the smallest variance. As shown in Table 6, based on the component numbering in Section 5.1, the optimal sequence was output, with a total objective value of 14.44, demonstrating excellent performance across all objectives. The model runtime was 58.46 s, with 960 parameters, reflecting high computational efficiency. The non-dominated solution ratio (0.042) and hypervolume (928) indicate good coverage in the objective space, further confirming the outstanding optimization results. All of these metrics fall within the 95 % confidence interval, suggesting the model has high stability and reliability, further supporting the effectiveness of the optimization.

4.4.2. Verification of algorithm effectiveness

In this experiment, we compared three task sequence optimization strategies: the disassembly cost minimum sequence, the maximum reliability sequence, and the balanced sequence generated by the multi-objective optimization algorithm. As shown in Table 7, the disassembly cost minimum sequence and the maximum reliability sequence only optimize a single objective without considering other objectives. In contrast, the multi-objective optimization algorithm achieves a total

Table 8
Algorithm Comparison.

Algorithm	TOV	RT/s	MPC	$S_{\text{non-dominated}}$	HV
NSGA-II	15.28	71.67	960	0.031	783
NSGA-III	14.93	86.51	960	0.035	637
SMS-EMOA	14.99	100.30	960	0.030	792
SPEA2	15.49	76.06	960	0.034	788
THG-MOEA/D	14.44	58.46	960	0.042	928

objective value of 14.44 by integrating and balancing various objectives, significantly outperforming the single-objective optimization results. This demonstrates that the multi-objective optimization method achieves better overall performance across multiple dimensions.

For the optimization of the battery disassembly process, the objectives include disassembly cost, carbon emissions, risk, and reliability. The distribution of all solutions obtained through the multi-objective optimization algorithm is shown in Fig. 10. The figure displays the distribution of all feasible solutions and the Pareto frontier solutions, with color and size encoding representing the reliability values of the solutions. The gradient from green to yellow indicates the change in reliability from low to high, visually illustrating the distribution of optimization results for disassembly performance under multi-objective trade-offs.

4.4.3. Algorithm comparison

In the data section, based on the combination of TransH and GIN, Table 8 compares five different optimization algorithms, including NSGA-II (Deb et al., 2002), NSGA-III (Deb and Jain, 2014), SMS-EMOA (Beume et al., 2007), SPEA2 (Zitzler et al., 2001), and the MOEA/D optimization algorithm used in this study's THG-MOEA/D model. NSGA-II focuses on fast non-dominated sorting to improve search efficiency; NSGA-III enhances optimization capabilities for high-dimensional objectives; SMS-EMOA improves solution coverage through hypervolume optimization criteria; while SPEA2 enhances solution diversity and quality through elitist strategies. By using the same

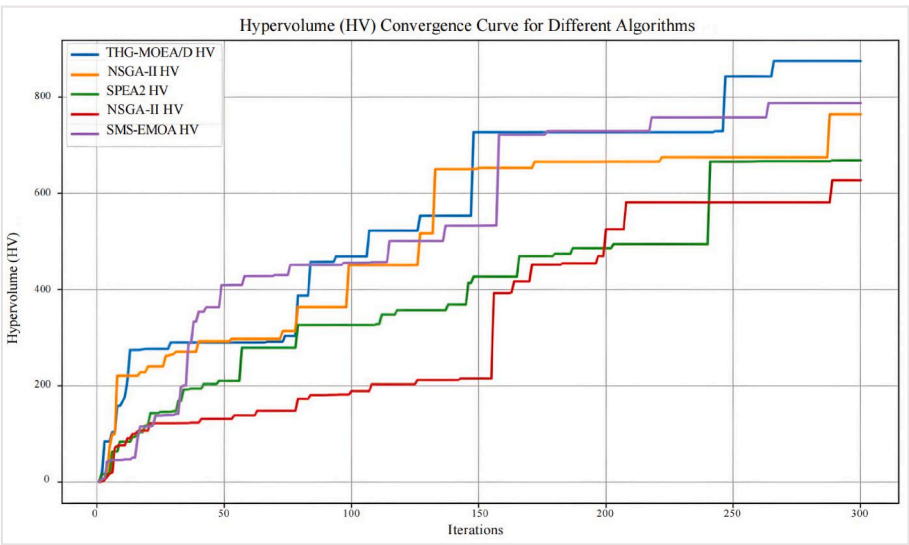


Fig. 11. Hypervolume Iteration Chart shows the progression of hypervolume values throughout the optimization process, illustrating how the hypervolume expands as the solutions move closer to the Pareto frontier, indicating improvements in solution quality.

Table 9
Model comparison.

Model	TOV	RT/s	MPC	$S_{\text{non-dominated}}$	HV
S1	17.38	83.42	960	0.029	815
S2	17.05	67.07	5920	0.032	816
S3	17.33	78.04	944	0.033	573
S4	16.87	85.80	944	0.030	620
S5	17.20	78.32	5904	0.036	515
S6	16.66	61.61	2760	0.033	768
S7	17.01	85.72	656	0.030	806
S8	16.82	72.26	1984	0.033	903
Ours	14.44	58.46	960	0.042	928

model parameters, the performance of different algorithms in multi-objective optimization is compared to assess the overall performance and relative advantages of THG-MOEA/D.

As shown in Table 8, the THG-MOEA/D model exhibits significant advantages across multiple key performance indicators. In terms of total objective value, THG-MOEA/D outperforms other algorithms,

demonstrating stronger global optimization capability. Additionally, THG-MOEA/D significantly reduces computation time, proving its higher efficiency. Moreover, THG-MOEA/D surpasses other comparison algorithms in both non-dominated solution ratio and hypervolume metrics, indicating a more balanced performance in solution diversity and objective space coverage. Overall, THG-MOEA/D outperforms traditional multi-objective optimization algorithms in both optimization effectiveness and computational performance.

Based on the aforementioned model, we demonstrate the convergence process of hypervolume (HV) under a maximum iteration count of 300, as shown in Fig. 11. It is evident that the MOEA/D used in this study rapidly improves during the early iterations and maintains a stable convergence trend in subsequent iterations, ultimately achieving the highest hypervolume value. Although NSGA-III and SPEA2 exhibit slightly slower convergence speeds, their HV values gradually approach higher levels as the number of iterations increases. In contrast, NSGA-II and SMS-EMOA have slower convergence rates, and their final HV values are significantly lower than those of the other algorithms, indicating weaker coverage in the objective space. Overall, MOEA/D

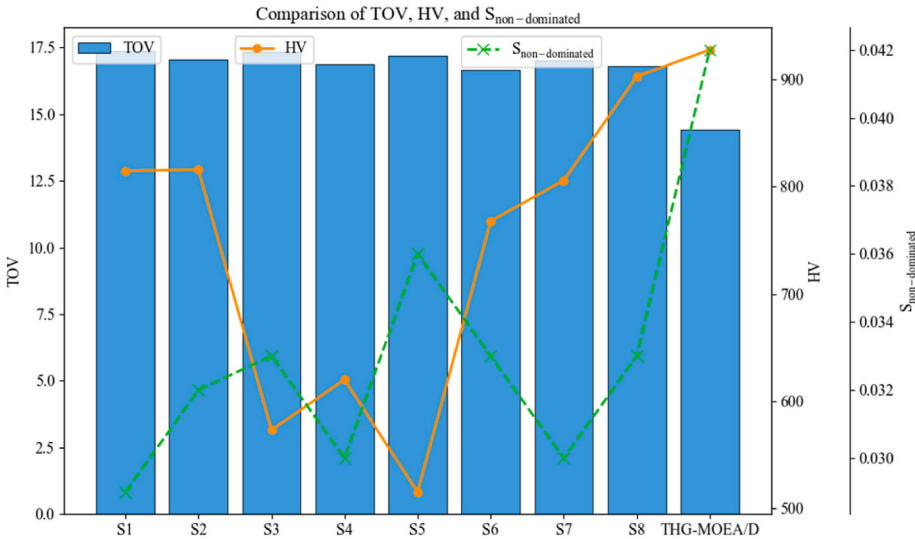


Fig. 12. Model Comparison Metrics Chart presents the comparison of different models based on key performance metrics, highlighting their relative strengths and weaknesses in the process.

demonstrates stronger advantages in both convergence speed and final optimization results.

4.4.4. Model comparison

According to the THG-MOEA/D model integrated in this study, we conducted a comparative analysis of various models, including TransE, TransR (Lin et al., 2015), GCN, and Graph Attention Network (GAT) (Veličković et al., 2018), as shown in Table 9. The subsequent optimization algorithm employed is MOEA/D, and the evaluation is performed across multiple dimensions such as Total Objective Value (TOV), Model Complexity (PC), and Runtime (RT). The models compared include S1 (TransH + GCN), S2 (TransH + GIN), S3 (TransE + GIN), S4 (TransE + GCN), S5 (TransE + GAT), S6 (TransR + GIN), S7 (TransR + GCN), and S8 (TransR + GAT). The results indicate that the THG-MOEA/D model (Ours) demonstrates optimal performance in terms of total objective value (TOV), achieving a value of 14.44, which is significantly superior to other models, reflecting its exceptional performance in multi-objective balanced optimization.

As shown in Table 9, while S8 performs well in hypervolume (HV) with a value of 903, indicating certain advantages in solution space exploration, its total objective value remains lower than that of the THG-MOEA/D model in this study. This suggests that there are shortcomings in the trade-offs between objectives. In terms of model parameter complexity (MPC), the number of parameters for Ours is 960, which is at a moderate level compared to other models, particularly when contrasted with the high parameter counts of S2 (5920) and S5 (5904), making Ours more efficient. Additionally, Ours also outperforms most models in runtime, showcasing high computational efficiency. Overall, the THG-MOEA/D model exhibits significant advantages in multi-objective optimization performance, model complexity, and computational efficiency.

As shown in Fig. 12, the graph generated based on Table 9 clearly illustrates the comparison results for Total Objective Value (TOV), Hypervolume (HV), and Non-Dominated Solution Ratio (Snon-dominated). It is evident that the THG-MOEA/D model performs best in TOV, achieving the lowest objective value and optimal optimization results. Additionally, it demonstrates high diversity in the non-dominated solution ratio, while S8, despite performing well in hypervolume, still falls short of THG-MOEA/D in TOV. Overall, THG-MOEA/D exhibits significant advantages across multiple metrics.

5. Conclusion

The THG-MOEA/D model proposed in this paper combines TransH knowledge graph embedding, Graph Isomorphism Network (GIN), and the decomposition-based multi-objective optimization algorithm (MOEA/D) to effectively address the multi-objective optimization problem in the lithium-ion battery disassembly process. By integrating TransH and GIN, the model captures the complex relationships between components and tools during disassembly and enhances its ability to learn the topological structure of the disassembly process. The introduction of the MOEA/D algorithm enables the model to achieve an optimal balance across multiple objectives, such as disassembly cost, carbon emissions, and risk, generating an efficient Pareto front solution set. Experimental results demonstrate that THG-MOEA/D outperforms classical algorithms (such as NSGA-II, SPEA2, and SMS-EMOA) in terms of total objective value (TOV), solution set coverage, and the proportion of non-dominated solutions, confirming its advantages in multi-objective optimization.

The main contribution of this model is the novel application of knowledge graph embedding technology combined with graph neural networks in battery disassembly sequence optimization, filling the gap in existing research on multi-objective optimization. THG-MOEA/D provides a new optimization framework for the battery recycling industry, overcoming the trade-offs between recovery efficiency, cost control, and environmental concerns in traditional disassembly

methods. However, as the scale of the battery model and the complexity of the disassembly process increase, the computational resource consumption and running time also rise significantly. Future work could focus on optimizing the neighborhood search strategy and convergence mechanism of MOEA/D. Moreover, the application of THG-MOEA/D is not limited to battery disassembly but can be extended to other fields, such as electronic waste recycling and automotive disassembly. Future research could also introduce multimodal data to further enhance the model's optimization capabilities. This model provides new approaches for multi-objective optimization problems, with broad application potential, particularly in intelligent manufacturing and resource management fields.

CRedit authorship contribution statement

Lifeng Yin: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chuang Wang:** Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization. **Liang Cong:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Quanyi Du:** Software, Resources, Methodology, Investigation, Formal analysis, Data curation.

Declaration of competing interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Data availability

Data will be made available on request.

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